

Fault Detection Electrical Models

Comparative Evaluation Using MLflow

Technical Evaluation Report

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This report presents a comparative analysis of machine learning models for photovoltaic fault detection using electrical features and MLflow-based performance evaluation.

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1 Overall Model Comparison

This section compares the overall performance of all evaluated models using standard classification metrics.

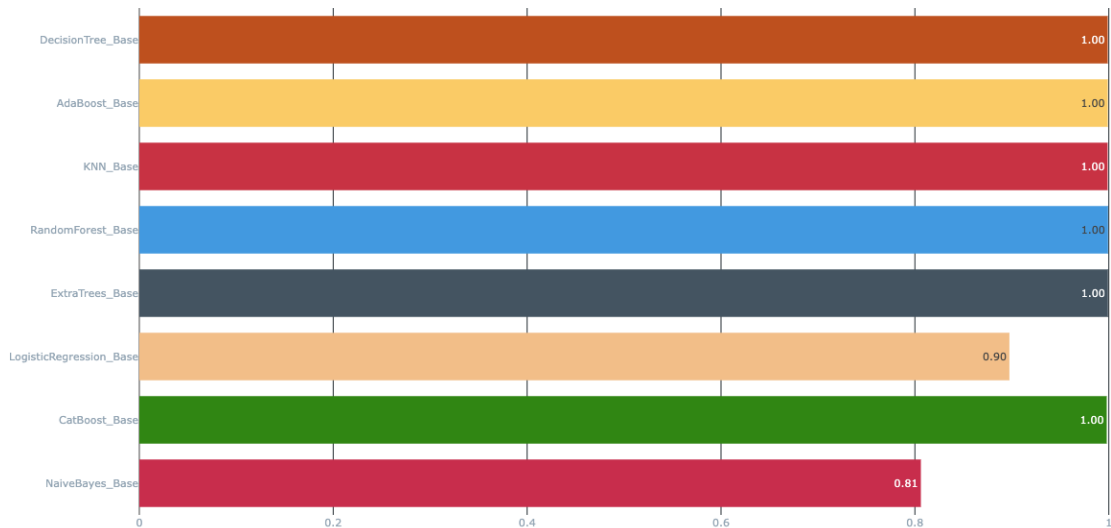


Figure 1: Comparison of Accuracy Across Models

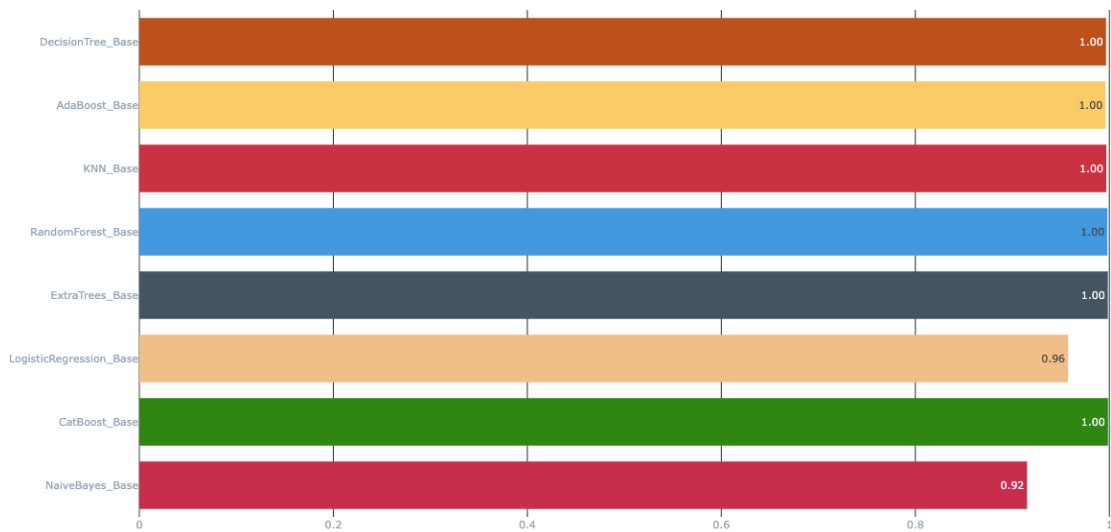


Figure 2: Comparison of Balanced Accuracy Across Models

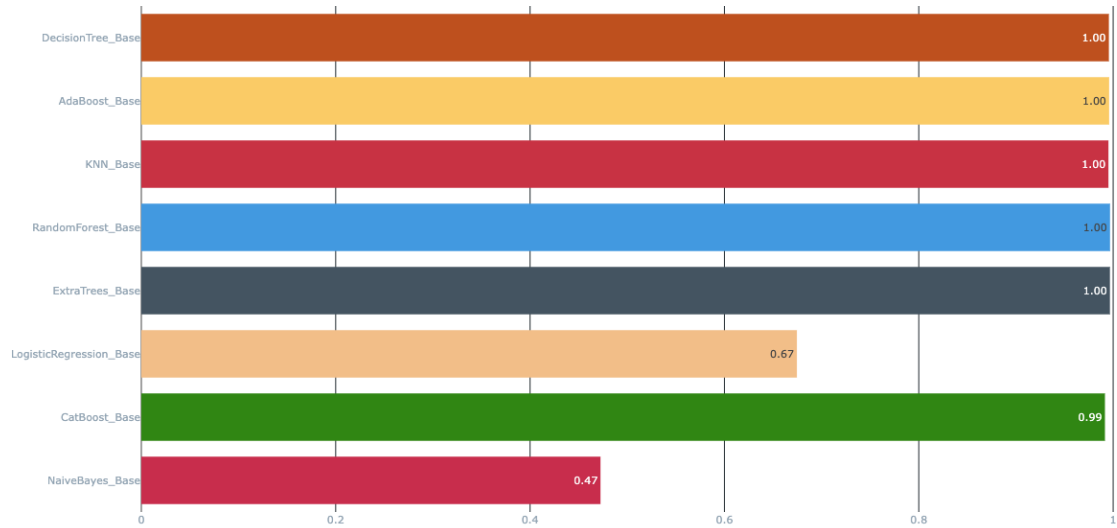


Figure 3: Comparison of Cohen's Kappa Across Models

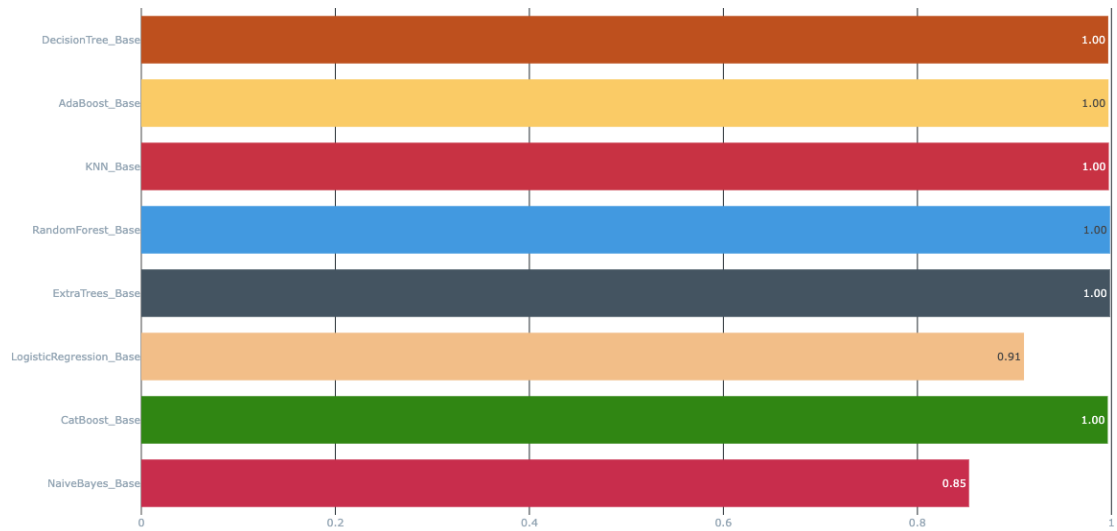


Figure 4: Comparison of F1-Score (Macro) Across Models

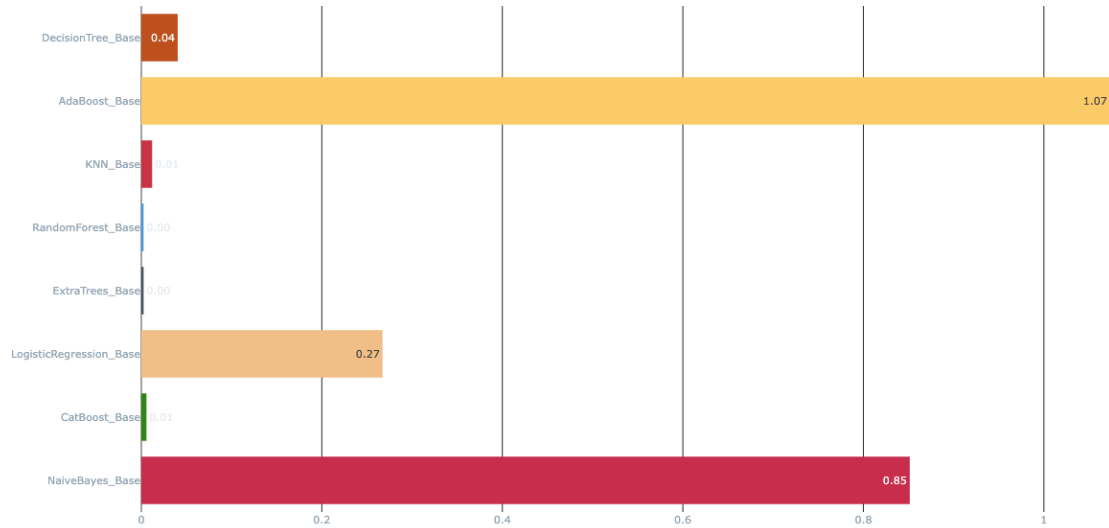


Figure 5: Comparison of Log Loss Across Models

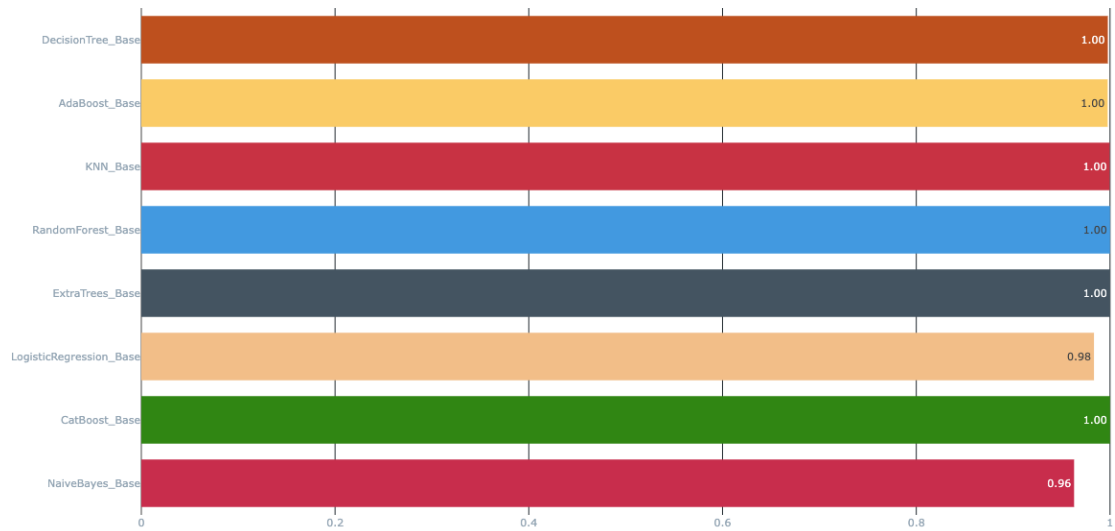


Figure 6: Comparison of ROC-AUC (Macro) Across Models

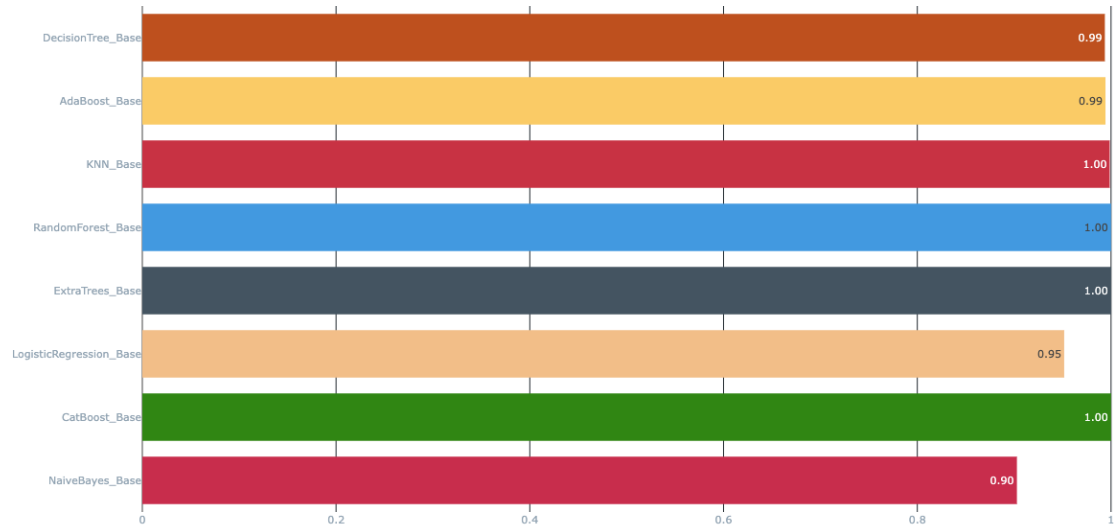


Figure 7: Comparison of PR-AUC (Macro) Across Models

2 Detailed Model Results

2.1 AdaBoost

Table 1: AdaBoost: Evaluation Metrics

Metric	Value
Accuracy	0.999
Balanced Accuracy	0.996
Precision (Macro)	0.998
Recall (Macro)	0.996
F1 Score (Macro)	0.997
Cohen's Kappa	0.996
Log Loss	1.073
ROC-AUC (Macro)	0.998
PR-AUC (Macro)	0.994

Table 2: AdaBoost: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	1.000	1.000	1.000	232587
Open Circuit	1.000	1.000	1.000	1205
Shadowing	1.000	1.000	1.000	37694
Short-Circuit	1.000	0.990	0.990	1200
Accuracy			1.000	272686
Macro Avg	1.000	1.000	1.000	272686
Weighted Avg	1.000	1.000	1.000	272686

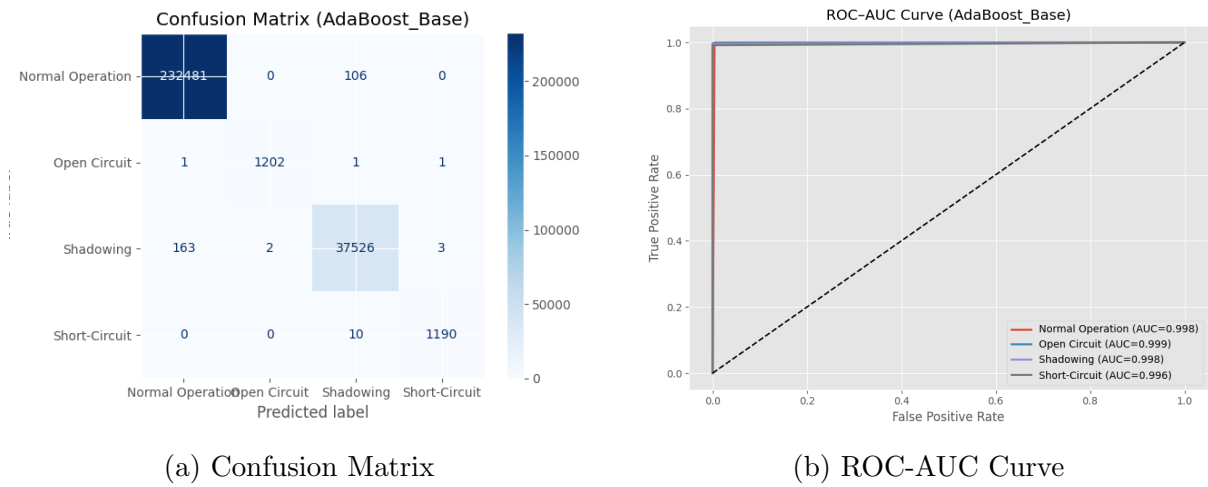


Figure 8: AdaBoost: Classification Performance

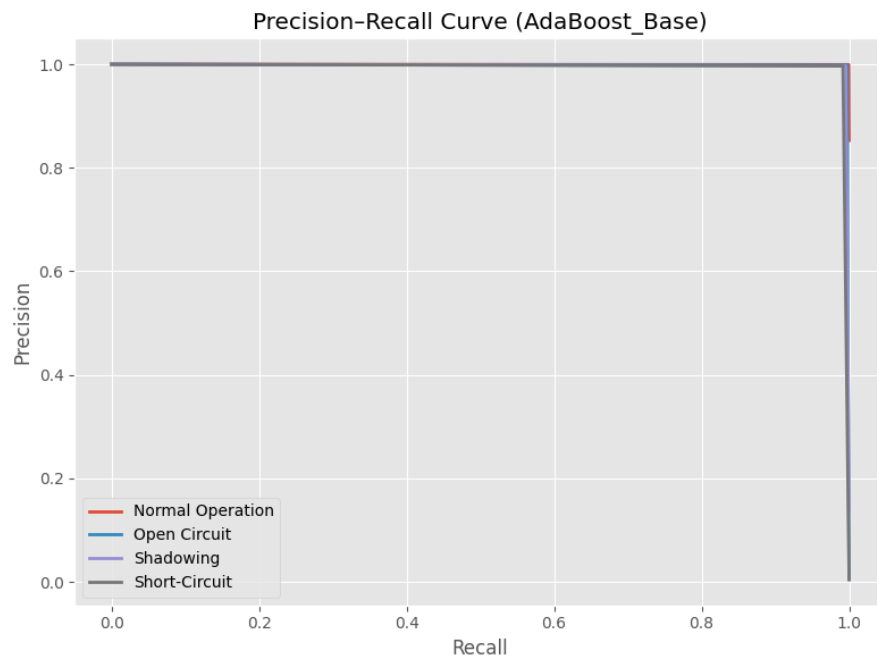


Figure 9: AdaBoost: Precision–Recall Curve

2.2 CatBoost

Table 3: CatBoost: Evaluation Metrics

Metric	Value
Accuracy	0.998
Balanced Accuracy	0.999
Precision (Macro)	0.994
Recall (Macro)	0.999
F1 Score (Macro)	0.996
Cohen's Kappa	0.992
Log Loss	0.006
ROC-AUC (Macro)	1.000
PR-AUC (Macro)	1.000

Table 4: CatBoost: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	1.000	1.000	1.000	232587
Open Circuit	1.000	1.000	1.000	1205
Shadowing	0.990	1.000	0.990	37694
Short-Circuit	0.990	1.000	0.990	1200
Accuracy			1.000	272686
Macro Avg	0.990	1.000	1.000	272686
Weighted Avg	1.000	1.000	1.000	272686

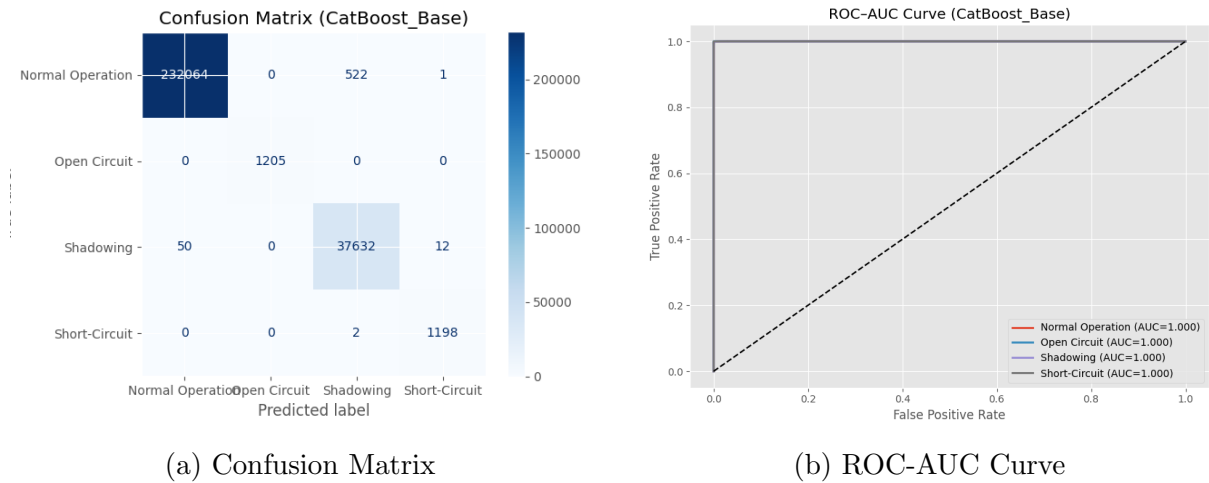


Figure 10: CatBoost: Classification Performance

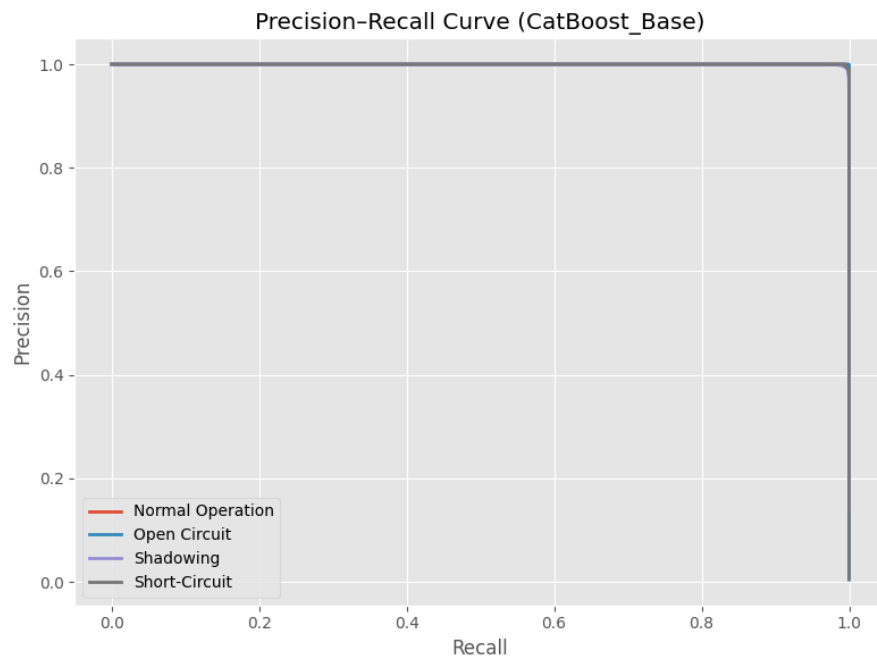


Figure 11: CatBoost: Precision-Recall Curve

2.3 K-Nearest Neighbours (KNN)

Table 5: KNN: Evaluation Metrics

Metric	Value
Accuracy	0.999
Balanced Accuracy	0.997
Precision (Macro)	0.998
Recall (Macro)	0.997
F1 Score (Macro)	0.997
Cohen's Kappa	0.995
Log Loss	0.012
ROC-AUC (Macro)	1.000
PR-AUC (Macro)	0.999

Table 6: KNN: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	1.000	1.000	1.000	232587
Open Circuit	1.000	1.000	1.000	1205
Shadowing	1.000	1.000	1.000	37694
Short-Circuit	1.000	0.990	0.990	1200
Accuracy			1.000	272686
Macro Avg	1.000	1.000	1.000	272686
Weighted Avg	1.000	1.000	1.000	272686

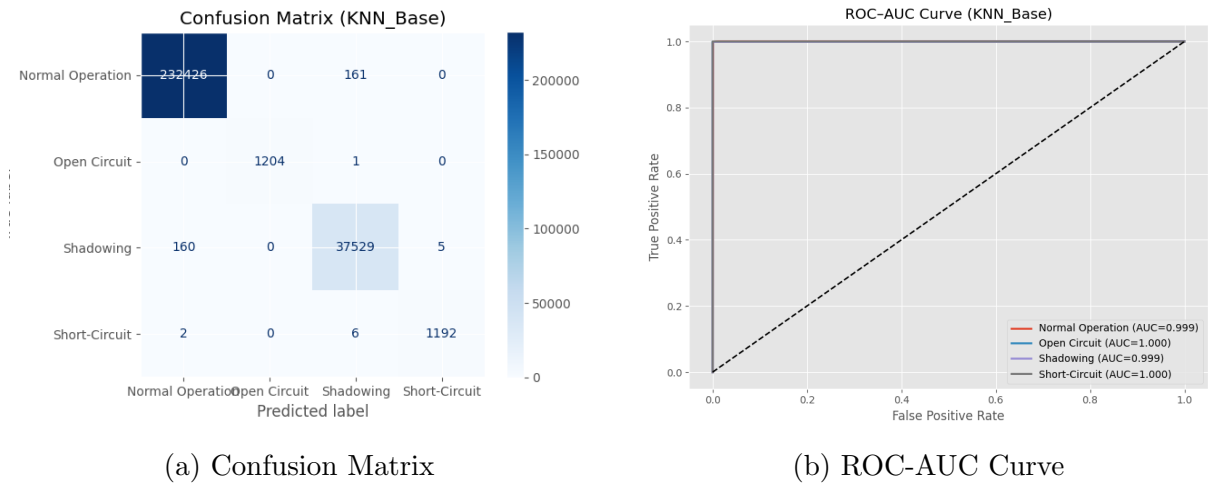


Figure 12: KNN: Classification Performance

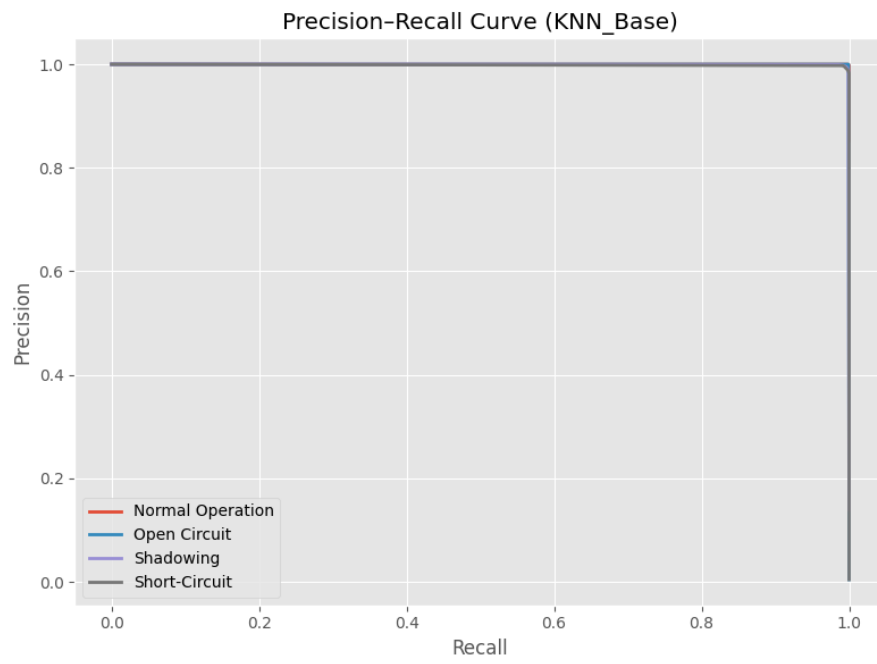


Figure 13: KNN: Precision-Recall Curve

2.4 Naive Bayes

Table 7: Naive Bayes: Evaluation Metrics

Metric	Value
Accuracy	0.806
Balanced Accuracy	0.915
Precision (Macro)	0.839
Recall (Macro)	0.915
F1 Score (Macro)	0.853
Cohen's Kappa	0.472
Log Loss	0.851
ROC-AUC (Macro)	0.963
PR-AUC (Macro)	0.903

Table 8: Naive Bayes: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	0.980	0.790	0.870	232587
Open Circuit	0.980	1.000	0.990	1205
Shadowing	0.410	0.880	0.560	37694
Short-Circuit	0.990	0.990	0.990	1200
Accuracy			0.810	272686
Macro Avg	0.840	0.920	0.850	272686
Weighted Avg	0.900	0.810	0.830	272686

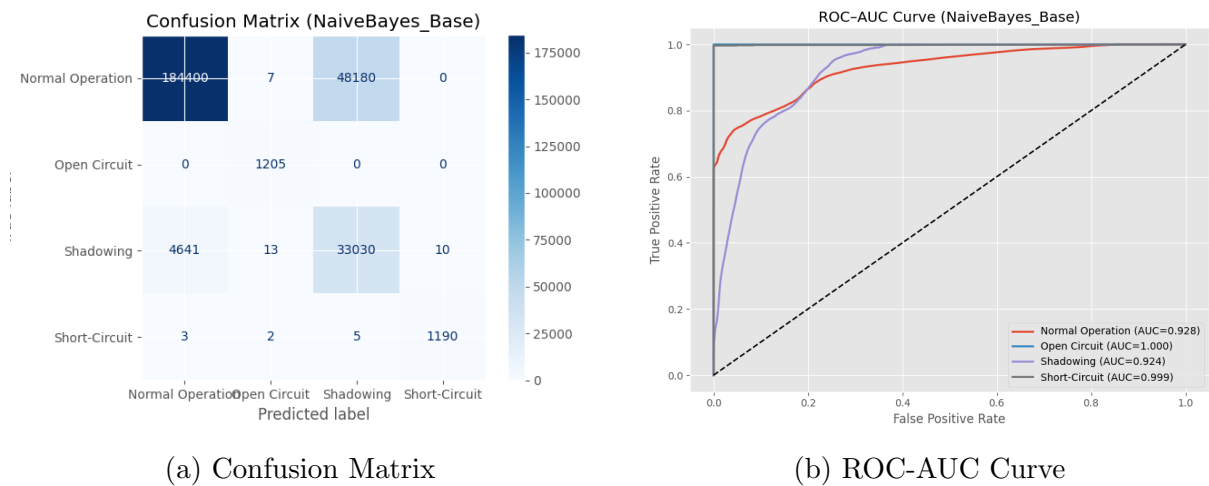


Figure 14: Naive Bayes: Classification Performance

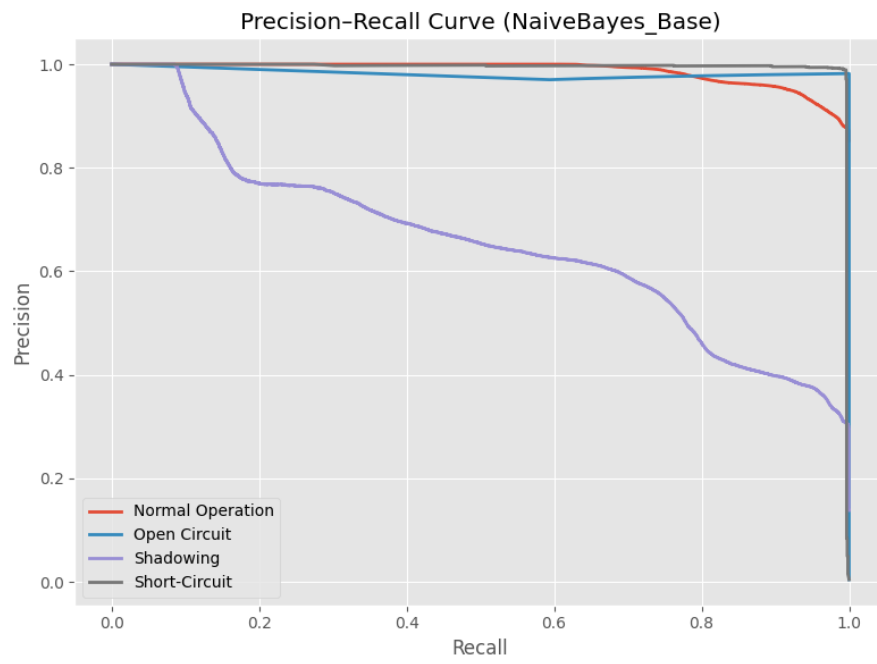


Figure 15: Naive Bayes: Precision-Recall Curve

2.5 Random Forest

Table 9: Random Forest: Evaluation Metrics

Metric	Value
Accuracy	0.999
Balanced Accuracy	0.998
Precision (Macro)	0.999
Recall (Macro)	0.998
F1 Score (Macro)	0.999
Cohen's Kappa	0.997
Log Loss	0.002
ROC-AUC (Macro)	1.000
PR-AUC (Macro)	1.000

Table 10: Random Forest: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	1.000	1.000	1.000	232587
Open Circuit	1.000	1.000	1.000	1205
Shadowing	1.000	1.000	1.000	37694
Short-Circuit	1.000	1.000	1.000	1200
Accuracy			1.000	272686
Macro Avg	1.000	1.000	1.000	272686
Weighted Avg	1.000	1.000	1.000	272686

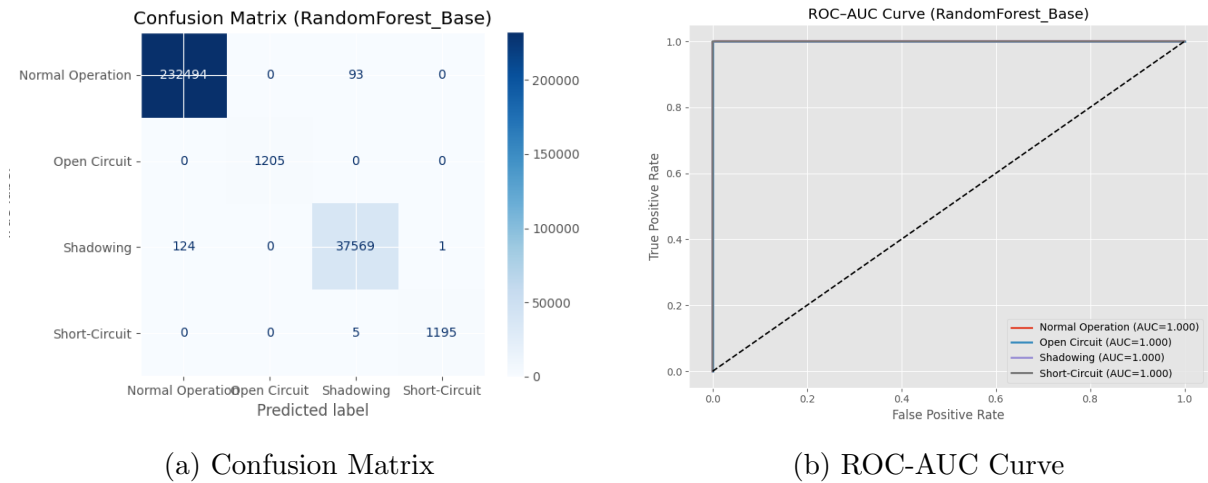


Figure 16: Random Forest: Classification Performance

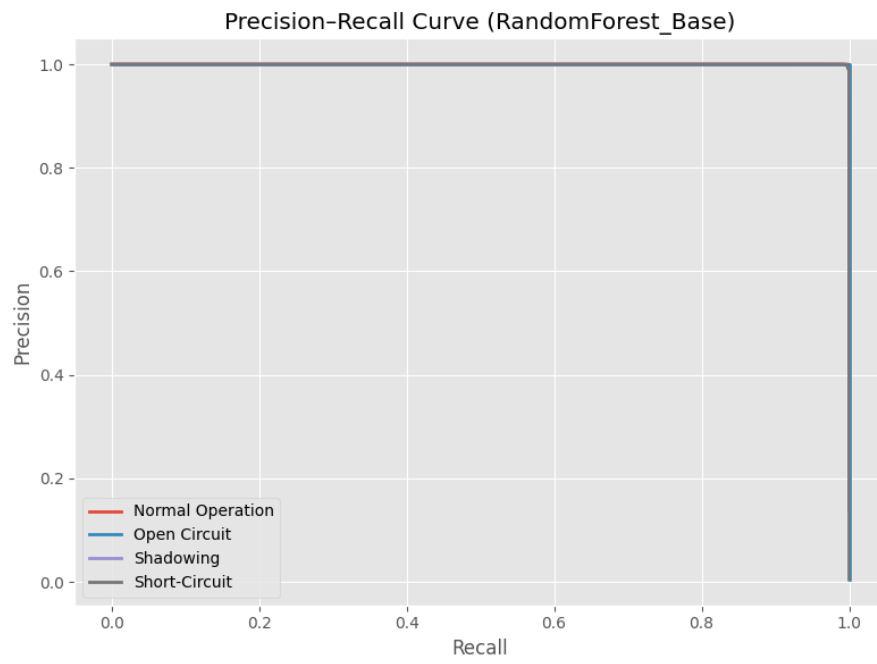


Figure 17: Random Forest: Precision-Recall Curve

2.6 Extra Trees

Table 11: Extra Trees: Evaluation Metrics

Metric	Value
Accuracy	0.999
Balanced Accuracy	0.998
Precision (Macro)	0.999
Recall (Macro)	0.998
F1 Score (Macro)	0.998
Cohen's Kappa	0.997
Log Loss	0.003
ROC-AUC (Macro)	1.000
PR-AUC (Macro)	1.000

Table 12: Extra Trees: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	1.000	1.000	1.000	232587
Open Circuit	1.000	1.000	1.000	1205
Shadowing	1.000	1.000	1.000	37694
Short-Circuit	1.000	1.000	1.000	1200
Accuracy			1.000	272686
Macro Avg	1.000	1.000	1.000	272686
Weighted Avg	1.000	1.000	1.000	272686

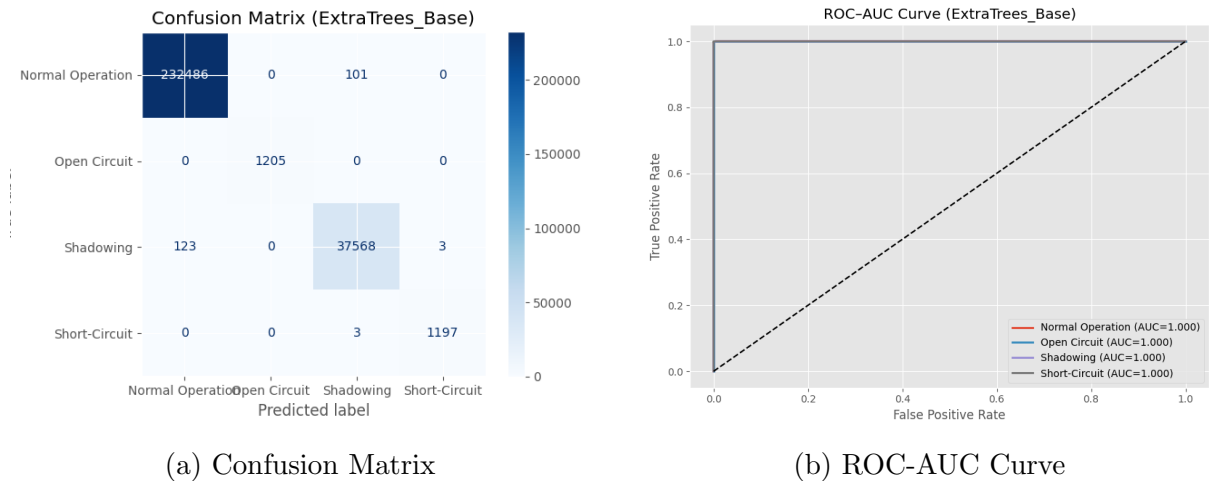


Figure 18: Extra Trees: Classification Performance

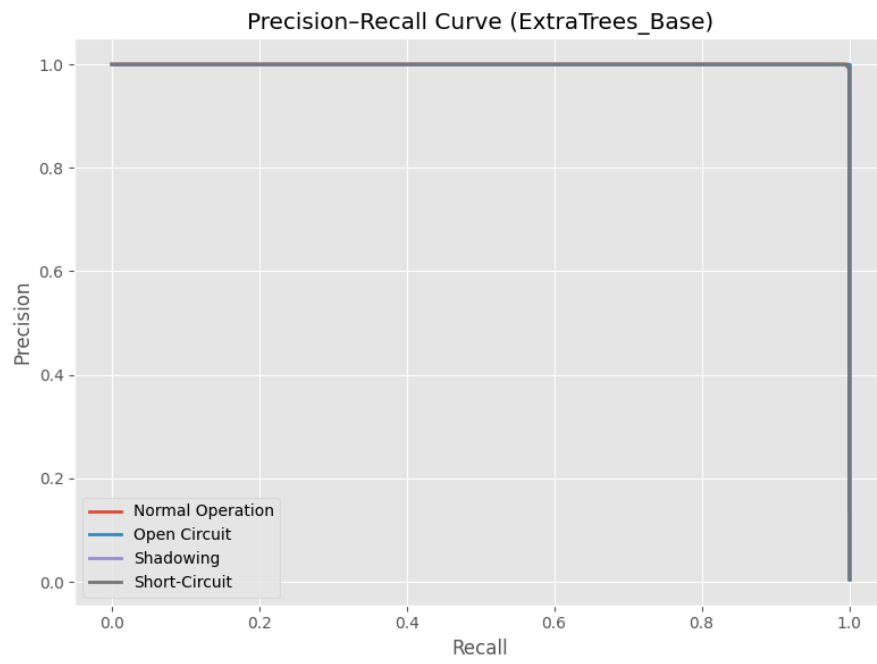


Figure 19: Extra Trees: Precision-Recall Curve

2.7 Logistic Regression

Table 13: Logistic Regression: Evaluation Metrics

Metric	Value
Accuracy	0.897
Balanced Accuracy	0.957
Precision (Macro)	0.885
Recall (Macro)	0.957
F1 Score (Macro)	0.910
Cohen's Kappa	0.675
Log Loss	0.267
ROC-AUC (Macro)	0.983
PR-AUC (Macro)	0.952

Table 14: Logistic Regression: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	0.990	0.890	0.940	232587
Open Circuit	1.000	1.000	1.000	1205
Shadowing	0.580	0.940	0.720	37694
Short-Circuit	0.970	1.000	0.990	1200
Accuracy			0.900	272686
Macro Avg	0.890	0.960	0.910	272686
Weighted Avg	0.930	0.900	0.910	272686

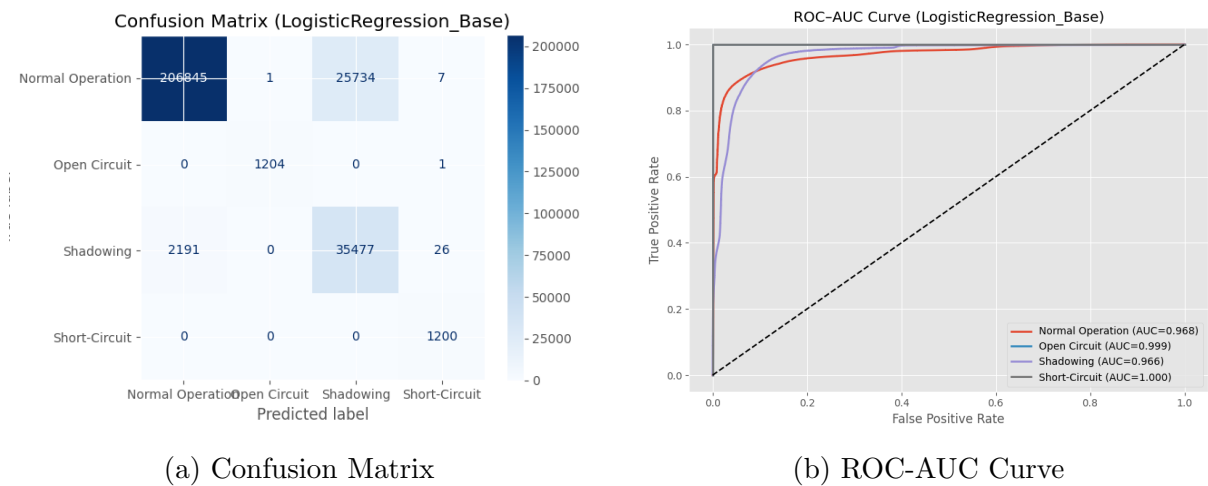


Figure 20: Logistic Regression: Classification Performance

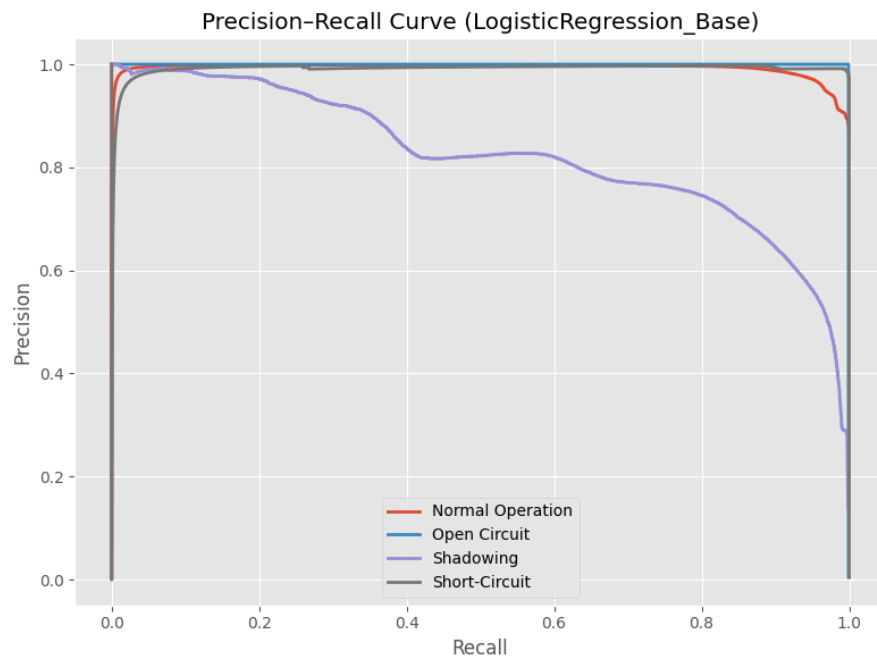


Figure 21: Logistic Regression: Precision-Recall Curve

2.8 Decision Tree

Table 15: Decision Tree: Evaluation Metrics

Metric	Value
Accuracy	0.999
Balanced Accuracy	0.997
Precision (Macro)	0.997
Recall (Macro)	0.997
F1 Score (Macro)	0.997
Cohen's Kappa	0.996
Log Loss	0.040
ROC-AUC (Macro)	0.998
PR-AUC (Macro)	0.994

Table 16: Decision Tree: Classification Report

Class	Precision	Recall	F1	Support
Normal Operation	1.000	1.000	1.000	232587
Open Circuit	1.000	1.000	1.000	1205
Shadowing	1.000	1.000	1.000	37694
Short-Circuit	1.000	0.990	0.990	1200
Accuracy			1.000	272686
Macro Avg	1.000	1.000	1.000	272686
Weighted Avg	1.000	1.000	1.000	272686

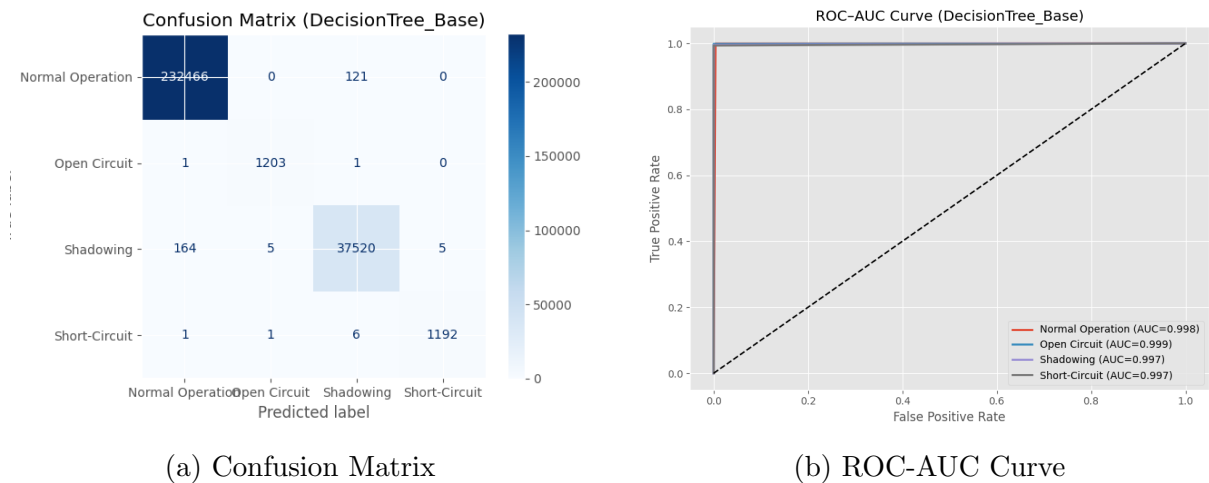


Figure 22: Decision Tree: Classification Performance

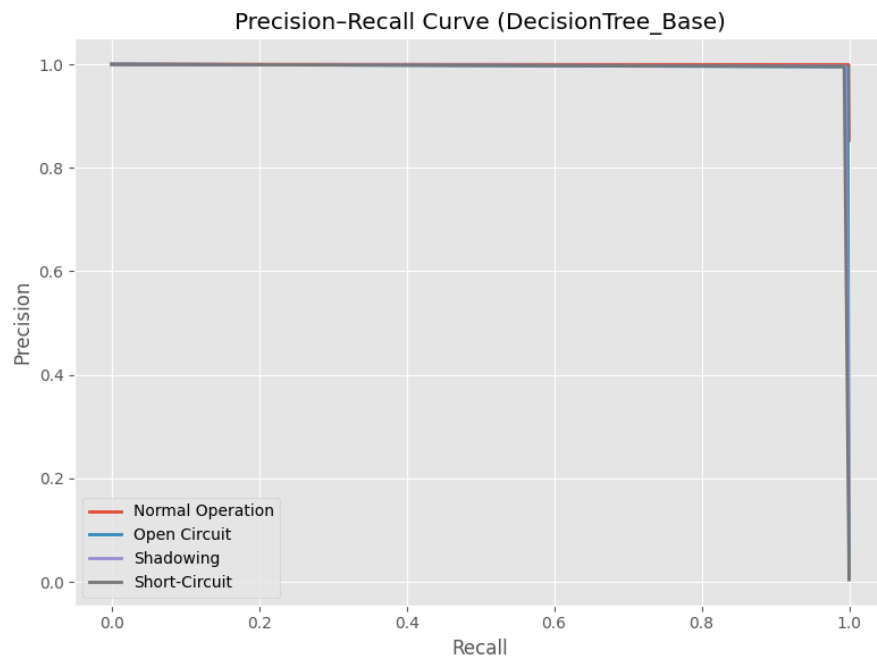


Figure 23: Decision Tree: Precision–Recall Curve